

Relate Fractions, Decimals, and Percents

Lesson 4-2

Name: _____

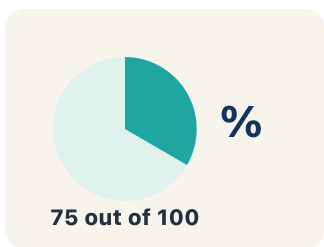
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

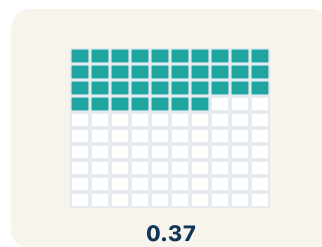
Picture first, then the word, then a plain-language meaning. Say each word out loud.



25% means 25 out of 100

Percent

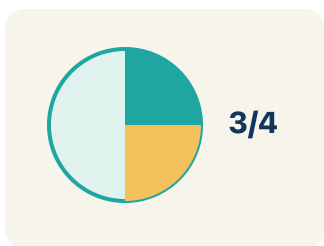
Write the definition:



0.75 means seventy-five hundredths

Decimal

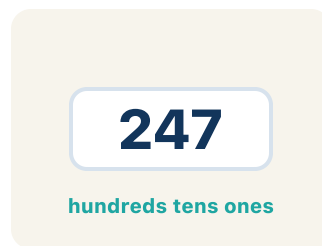
Write the definition:



$1/2 = 0.5 = 50\%$

Equivalent

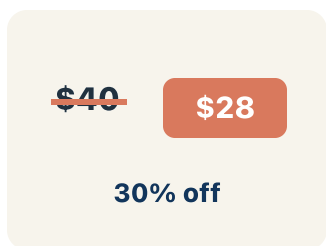
Write the definition:



Common benchmarks: 25%, 50%, 75%

Benchmark

Write the definition:



\$40 shirt at 25% off: discount = \$10

Discount

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can write equivalent fractions, decimals, and percents for the same value.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Percent

Decimal

Equivalent

Benchmark

Discount

1

A ratio that compares a number to 100, written with a % sign, is a _____.

2

A number that uses a point to show parts of a whole is a _____.

3

_____ — Having the same value, just written a different way.

4

A common, easy-to-remember value like 50% or $\frac{1}{4}$ used for comparing is a

_____.

5

An amount taken off a price, often given as a percent, is a _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Convert $\frac{3}{5}$ to a percent.

- A. 60%
- B. 35%
- C. 53%
- D. 65%

Step 1 $3 \div 5 = 0.6 = 60\%$.

 **Answer:** A. 60%

 We do — together

Solve this one with your class using the same steps.

Problem: Convert $\frac{7}{20}$ to a percent.

- A. 35%
- B. 70%
- C. 20%
- D. 14%

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which decimal is equivalent to 45%?

- A. 0.45
- B. 4.5
- C. 0.045
- D. 45.0

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Your arcade scoreboard reads 0.25 of the level cleared. What percent and fraction match this score?

- A. 25% and $\frac{1}{4}$
- B. 2.5% and $\frac{1}{4}$
- C. 25% and $\frac{1}{5}$
- D. 250% and $\frac{1}{40}$

Show your work:

2. A player powered up $\frac{3}{5}$ of the energy bar. Which decimal and percent name the same amount?

- A. 0.6 and 60%
- B. 0.35 and 35%
- C. 0.6 and 6%
- D. 0.65 and 65%

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student says $\frac{3}{8}$ is greater than 40% because 3 and 8 are bigger numbers than 4 and 0. Is the student correct? Convert both values to the same form and explain.

Sentence starter: The student is ____ because $\frac{3}{8} = \frac{\quad}{\quad} = \quad\%$, which is ____ than 40%.

Show your work:

Reflect — Exit Ticket

Which shows 0.45 as a fraction and a percent?

- A. $\frac{9}{20}$ and 45%
- B. $\frac{45}{10}$ and 4.5%
- C. $\frac{4}{5}$ and 45%
- D. $\frac{9}{20}$ and 4.5%

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Percent
2. **Notes 2:** Decimal
3. **Notes 3:** Equivalent
4. **Notes 4:** Benchmark
5. **Notes 5:** Discount
6. **We Try (notes):** A. $35\% - 7 \div 20 = 0.35 = 35\%$.
7. **You Try (notes):** A. 0.45 — *To convert a percent to a decimal, divide by 100: $45 \div 100 = 0.45$.*
8. **Try It 1:** A. 25% and $1/4$ — *Move the decimal two places: $0.25 = 25\%$. As a fraction, $0.25 = 25/100 = 1/4$.*
9. **Try It 2:** A. 0.6 and 60% — *Divide $3 \div 5 = 0.6$, then multiply by 100: $0.6 = 60\%$. So $3/5 = 0.6 = 60\%$.*
10. **Exit Ticket:** A. $9/20$ and 45% — $0.45 = 45/100 = 9/20$ (simplified). $0.45 \times 100 = 45\%$.